# How Is Gender Portrayed on Preschool Children's Book Covers? An Analysis of the Chinese National Library Catalogue between 2012-2022

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1. Background

The multimodality, especially the visual elements of picture books, empowers preschool children to read and perceive gender and other features independently (Danko-McGhee & Slutsky, 2011; Jalongo, 2004) . Research suggests preschool children are mostly illiterate, but they can recognise stories or books from covers and illustrations, they are even able to search for books in the libraries by describing the book cover (Danko-McGhee & Slutsky, 2011; Kemple & Lopez, 2009; Li, 2022) . Therefore, gender representations in these titles would influence how children see and think of gender by emphasising characters, narratives and storytelling (Bal & Veltkamp, 2013; Bartholomaeus, 2016; Caple & Tian, 2022; Faver & Alanis, 2012) . Library catalogues are filtered by librarians annually, and book selections are usually reviewed by experts in the children's literature field. Library is a place to assist children's right to search for information (Brock & Rankin, 2012, p. 75; Busayo, 2011) , library catalogues should be inclusive and cover different aspects of life, including gender. Book covers and titles, aside from the main body, are important parts of peritext (Genette, 1997) and can reveal the beginning of storytelling in preschool books (Solomon Eaglstein & Berman, 1990) . Previous studies examined book covers focus on their functions for books and diverse styles and designing (Gruppetta, 2008; Osvath, 2021) , but lack analysis from a gender perspective based on digital methods. To merge the gap, this study applies a combination of ChatGPT and text mining on the Chinese children's library catalogues with a 10-year timeline (2012-2022).

1. Method
   1. Data Collection

We collected the children's book catalogue of the National Library of China between 2012 and 2022[1] and selected books that target 0-6-year-old young readers as our research objects. We then searched ISBN for all titles on Douban (the largest reading community in China, equalise Goodreads)[2] , and crawled book covers of all 6,229 titles to make up the initial dataset using Python.

1. Data process

We first uploaded book covers of 6,229 titles one by one. In the initial stage, we asked ChatGPT[3] two questions for each title: 1) Is gender inferred in the book title? If so, what is it? 2) Are there any other elements suggesting gender? If so, what are they?

For each title, we will get two independent textual answers judging gender expressions from the book title and illustrations on book covers. Answers are semi-structured, suggesting if the title and illustrations infer gender or not, with further explaining the gendered word/phrase. For example, in the answer for a book title "Yes, gender is inferred in the book title ##### ( *This Is My Dad* , Hout, 2022) . The term ## (dad) explicitly refers to a male parent, so the gender of the subject, "dad," is male"(ChatGPT 4.0, 2024), ChatGPT recognise the title includes a male character and identify the gender word "##dad". Similar answer structure are reported for the analysis of illustrations on book covers.

To check the precision of ChatGPT in gender detection, the two authors manually coded the first 60 books and compared them with ChatGPT answers. The Kappa coefficient for inter-rater reliability was 0.82, suggesting substantial agreement between coders.

To follow up, ChatGPT answers will make up a final corpus for our further analysis. We will use the term frequency to analyse the recurrences of gender/non-gender titles, classify the uses of different elements in illustrations to portray gender, and see how they vary from time (publication date) and book genre (CLCS[4] ).

*Figure 1* Research Process: Gender Detection on Children's Library Catalogue of NLC

5.

1. Results

The results of this paper will be presented at the conference.

1. Implications

This study uses ChatGPT 4.0 to recognise gender embedded in preschool children's books which are displayed in Chinese public libraries. It will reveal the semantic networks and associations between words to imply gender in children's books. Based on the large language model, ChatGPT can exhibit gender perceptions for book covers from a public perspective, although it might also reflect gender biases (Alexopoulos, Lyons, Mahetaji, Barnes, & Gutwillinger, 2023; Gross, 2023; Sebo, 2024) and need to be improved. However, this will help librarians identify gender role assumptions and be aware of gender biases in categorising titles for future practice, thus improving gender equality and boosting diverse representations in the library collection of preschool children's books.

# Notes

1. The catalogue was published on the website of the National Library of China on 3/1/2024. Data can be publicly accessed at: http://kids.nlc.cn/srtsg/sytc/20240104_2637583.shtml
2. See https://www.douban.com to access Douban. Compared to WorldCat which is an open online platform and provides library archives and collections, Douban provides covers for Chinese books with high resolution.
3. ChatGPT supports Chinese and provides English analysis.
4. Chinese Library Classification System, see at https://www.clcindex.com

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